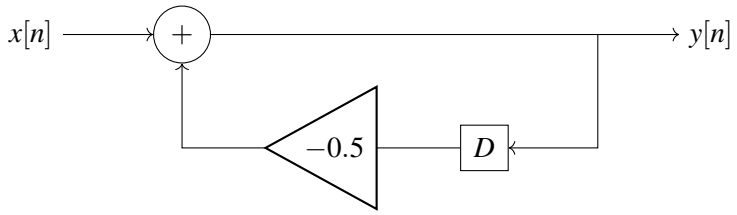
# EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 10B

**1. Frequency Response from a DAG Block Diagram [Adapted from EE 120]**

Consider a DT LTI system with input $x[n]$ and output $y[n]$ that is implemented by the following block diagram:




(a) Find the LCCDE that describes this system.

**Answer:** The difference equation describing this system is

$$y[n] = x[n] - 0.5y[n-1]$$

(b) Find the frequency response $H(\omega)$ of this system.

**Answer:** Using the eigenfunction property by setting $x[n] = e^{i\omega n}$ and $y[n] = H(\omega)e^{i\omega n}$,

$$\begin{aligned} H(\omega)e^{i\omega n} &= e^{i\omega n} - 0.5H(\omega)e^{i\omega(n-1)} \\ &= e^{i\omega n} - 0.5H(\omega)e^{-i\omega}e^{i\omega n} \\ \left(1+0.5e^{-i\omega}\right)e^{i\omega n}H(\omega) &= e^{i\omega n} \\ H(\omega) &= \frac{e^{i\omega n}}{\left(1+0.5e^{-i\omega}\right)e^{i\omega n}} \\ &= \frac{1}{1+0.5e^{-i\omega}} \end{aligned}$$

(c) Is this a high pass or low pass filter?

**Answer:** We can think about this in two ways:

**1. Geometric Approach:**

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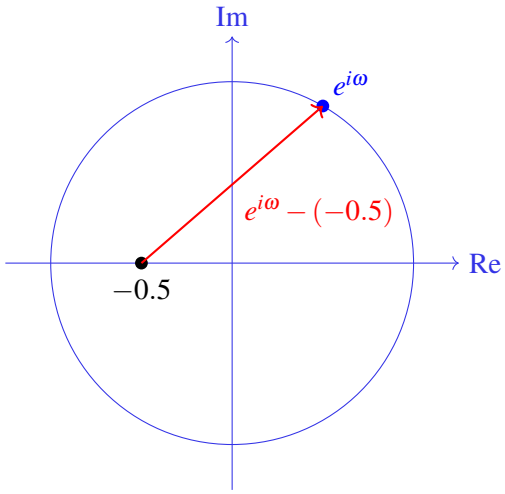
Recall from part (b) that the frequency response can be written as:

$$H(\omega) = \frac{1}{1+0.5e^{-i\omega}} = \frac{e^{i\omega}}{e^{i\omega}-(-0.5)}$$

The magnitude is therefore:

$$|H(\omega)| = \frac{1}{|e^{i\omega}-(-0.5)|}$$

This represents the reciprocal of the distance from the point $e^{i\omega}$ on the unit circle to the point located at $-0.5$ on the real axis:




Key observations:

- When $\omega = 0$, $e^{i\omega} = 1$. The distance from 1 to $-0.5$ is 1.5, so $|H(0)| = 1/1.5 \approx 0.667$.
- When $\omega = \pi$, $e^{i\omega} = -1$. The distance from $-1$ to $-0.5$ is 0.5, so $|H(\pi)| = 1/0.5 = 2$.
- As $\omega$ moves from 0 to $\pi$, the point $e^{i\omega}$ travels along the upper half of the unit circle from $(1,0)$ to $(-1,0)$. The distance to the point at $-0.5$ decreases monotonically, so $|H(\omega)|$ increases. Thus, we have a high-pass filter!

**2. Algebraic Approach:**

To determine the filter type, we simplify the magnitude of the frequency response first:

$$|H(\omega)| = \frac{1}{\sqrt{(1+0.5\cos\omega)^2+(0.5\sin\omega)^2}}$$

Expanding $(1+0.5\cos\omega)^2$ and $(0.5\sin\omega)^2$:

$$(1+0.5\cos\omega)^2 = 1^2+2(1)(0.5\cos\omega)+(0.5\cos\omega)^2 = 1+\cos\omega+0.25\cos^2\omega$$

$$(0.5\sin\omega)^2 = 0.25\sin^2\omega$$

Add the two results in the square root:

$$(1+0.5\cos\omega)^2+(0.5\sin\omega)^2 = \left[1+\cos\omega+0.25\cos^2\omega\right]+\left[0.25\sin^2\omega\right]$$

$$= 1+\cos\omega+0.25\left(\cos^2\omega+\sin^2\omega\right)$$

$$= 1 + \cos\omega + 0.25(1) = 1 + \cos\omega + 0.25 = 1.25 + \cos\omega$$

Substitute back into $|H(\omega)|$:

$$|H(\omega)| = \frac{1}{\sqrt{1.25 + \cos\omega}}$$

Now we analyze the magnitude at key frequencies:

Low frequency ($\omega = 0$):

$$|H(0)| = \frac{1}{\sqrt{1.25 + \cos 0}} = \frac{1}{1.5} \approx 0.667 \quad \text{(attenuated)}$$

High frequency ($\omega = \pi$):

$$|H(\pi)| = \frac{1}{\sqrt{1.25 + \cos\pi}} = 2 \quad \text{(passed)}$$

We can see that, again, the filter attenuates low frequencies and passes high frequencies, so it is a high-pass filter!

Both geometrically and algebraically, we conclude this is a **high-pass filter**. Nice!

## 2. 16A's Greatest Hits

Kevin is a DJ working on his newest audio filter, the Two Tap, which can be modelled as the following LCCDE:

$$y[n] = x[n] + x[n-2].$$

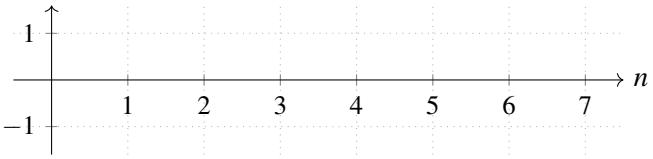
Kevin would like to mix the following beat using his audio filter:

$$x[n] = (-1)^n + \cos\left(\frac{\pi n}{2}\right),$$

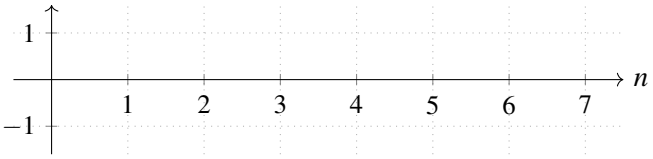
and would like to do so by passing it into the system.

(a) Kevin starts by trying to solve this by convolving the input signal. Sketch both components of the input signal below, then use your sketches and the system equation to find a closed-form expression for $y[n]$.

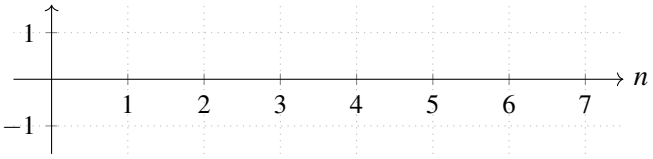
$h[n]$



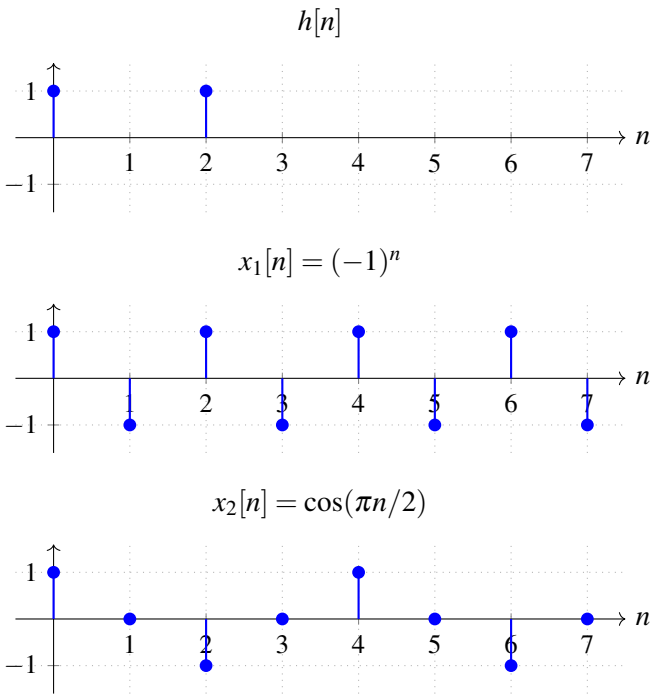

$x_1[n]$




$x_2[n]$




**Answer:**




We can compute the convolution for each component visually and then invoke linearity to add the outputs. Assume $x_1[n] = (-1)^n$, and $x_2[n] = \cos(\pi n/2)$

*Convolving $h$ with $x_1[n] = (-1)^n$:* From the plot, $x_1[n]$ has period 2, so shifting it by 2 samples produces an identical copy. The output is therefore $y_1[n] = x_1[n] + x_1[n-2] = (-1)^n + (-1)^n = 2(-1)^n$.

*Convolving $h$ with $x_2[n] = \cos(\pi n/2)$:* From the plot, $x_2[n]$ has period 4, so shifting by 2 samples lands exactly half a period later — every sample flips sign. The output is therefore $y_2[n] = x_2[n] + x_2[n-2] = x_2[n] - x_2[n] = 0$.

By linearity, the outputs add:

$$y[n] = y_1[n] + y_2[n] = 2(-1)^n + 0 = 2(-1)^n.$$

This is the output beat that Kevin sees as a result of passing his input through the Two Tap filter.

(b) Rival DJ Mallika sees Kevin running this computation, and would like to approach solving for the output to the Two Tap from a different perspective. To help Mallika, decompose $x[n]$ using a DTFS and identify the coefficients $X_k$ by inspection.

**Answer:** $x[n]$ is periodic with period $N = 4$, so the DTFS frequencies are $\omega_k = \frac{2\pi k}{N} = \frac{\pi k}{2}$ for $k = 0, 1, 2, 3$, and the synthesis equation is $x[n] = \sum_{k=0}^{3} X_k e^{i\omega_k n}$. Reading off each term of $x[n]$ directly:

- $(-1)^n = e^{i\pi n}$ sits at $\omega_2 = \pi$, so $X_2 = 1$.
- $\cos\left(\frac{\pi n}{2}\right) = \frac{1}{2}e^{i(\pi/2)n} + \frac{1}{2}e^{-i(\pi/2)n} = \frac{1}{2}e^{i\omega_1 n} + \frac{1}{2}e^{i\omega_3 n}$, so $X_1 = X_3 = \frac{1}{2}$.
- There is no DC component, so $X_0 = 0$.

(c) Suppose a general DT LTI system has frequency response $H(\omega)$, and the input $x[n]$ is periodic with DTFS coefficients $X_k$. Write a simple expression for the output $y[n]$.

**Answer:** The complex exponential $e^{i\omega n}$ is an eigenfunction of any LTI system:

$$e^{i\omega n} \xrightarrow{H} H(\omega) e^{i\omega n}.$$

Since $x[n] = \sum_k X_k e^{i\omega_k n}$, linearity gives:

$$y[n] = \sum_k X_k H(\omega_k) e^{i\omega_k n}.$$

Intuitively, this result means that, if we look at x as the *sum of its constituent frequencies*, and a system as only *scaling frequencies*, the output of a DT-LTI system to any arbitrary input can be thought of by *scaling each of the input's constituent frequencies by the frequency response.*

(d) Using the expression from part (c), help Mallika find $y[n]$ for this system. First determine $H(\omega_k)$, then apply it to the DTFS coefficients from part (b). Verify Mallika's answer matches up with Kevin's.

**Answer:** Since $e^{i\omega_k n}$ is an eigenfunction, the output satisfies $y[n] = H(\omega_k) e^{i\omega_k n}$. Substituting into the system equation:

$$H(\omega_k) e^{i\omega_k n} = e^{i\omega_k n} + e^{i\omega_k (n-2)},$$

so $H(\omega_k) = 1 + e^{-i2\omega_k}$.

Applying $y[n] = \sum_{k=0}^{3} H(\omega_k) X_k e^{i\omega_k n}$, we expand each term:

$$\begin{aligned}
k = 0: &\quad (1 + e^{0}) \cdot X_0 \cdot 1 = 2 \cdot 0 = 0 \\
k = 1: &\quad (1 + e^{-i\pi}) \cdot X_1 \cdot e^{i(\pi/2)n} = \underbrace{(1-1)}_{=0} \cdot \tfrac{1}{2} e^{i(\pi/2)n} = 0 \\
k = 2: &\quad (1 + e^{-i2\pi}) \cdot X_2 \cdot e^{i\pi n} = 2 \cdot 1 \cdot (-1)^n = 2(-1)^n \\
k = 3: &\quad (1 + e^{-i3\pi}) \cdot X_3 \cdot e^{i(3\pi/2)n} = \underbrace{(1-1)}_{=0} \cdot \tfrac{1}{2} e^{i(3\pi/2)n} = 0
\end{aligned}$$

Summing all terms:

$$y[n] = 0 + 0 + 2(-1)^n + 0 = 2(-1)^n.$$

This matches Kevin's solution! The $k = 1$ and $k = 3$ terms vanish because $e^{-i\pi} = -1$, making $H(\pi/2) = H(3\pi/2) = 0$ and annihilating both cosine components. On the other hand, $H(\pi) = 2$ doubles the $(-1)^n$ term.

Note that this is a general property: you can decompose any periodic signal into its DTFS coefficients, scale each basis vector by its frequency response, and use linearity to compute the output to the system. More extensions to this concepts in this vein are discussed in 16A's follow up *EE120*, such as through the perspective of the DTFT, a cousin to the DTFS.